Inaya, Jules and Zaiba Phase 1 March 10

Predicting Marine Biodiversity: A Comparison of Statistical Learning Methods

Problem Statement

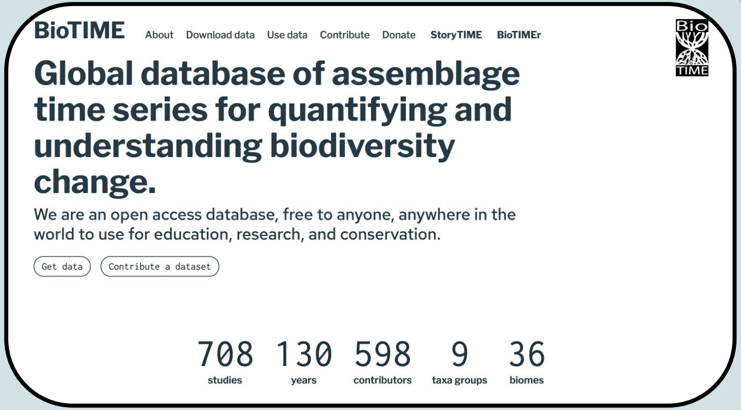
Question: What environmental factors predict species richness in marine ecosystems?

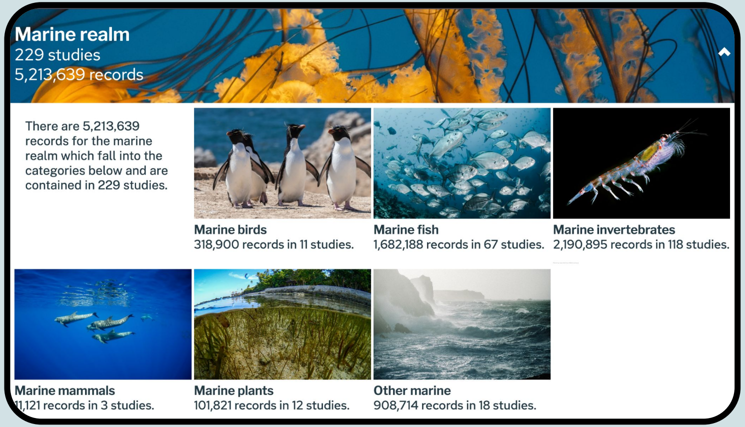
Approach: Compare Ridge/Lasso/Elastic Net, GAMs, and Random Forest

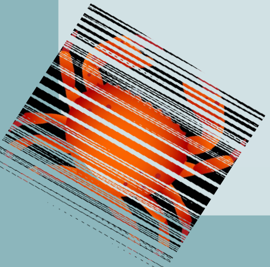
Why it matters: Understanding biodiversity patterns for conservation under climate change

Data

- ➔ **Dataset:** BioTIME 2.0 (temperate marine ecosystems)
- ➔ **Size:** 1,625 sites from 761K species records
- ➔ **Unit of analysis:** Individual sampling site with over 5 years of data (unique lat/lon location)
- ➔ **Target:** Species richness (count of species per site)
- ➔ **Features:** Geography (lat/lon, depth), climate proxies, human impact, sampling effort







Proposed Methods

Regularized Regression

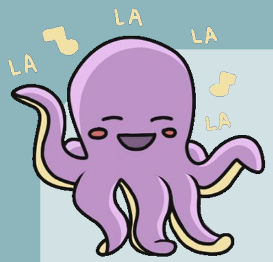
Goal:
- Handle correlated environmental predictors
- Identify important variables
- CV curve for α tuning

Generalized Additive Models

Goal:
- Capture nonlinear ecological relationships
- Estimate smooth curves for predictors
- Plots to view partial effect per predictor
- Wiggliness measure (EDF values)

Random Forest

Goal:
- Detect interactions
- Rank feature importance
- Find complex nonlinear patterns
- Partial dependence plots
- Max achievable R^2



Evaluation

- 5-fold cross validation, ensures models generalize beyond training data
- RMSE, prediction error magnitude
- R^2, explains variance

Comparison

- Which features consistently matter and show up in results?
- How much model improvement comes from each method (regularization, nonlinear relationships, interaction based)?

	RISKS	MITIGATIONS
1	**Sampling Bias** Species richness increases with sampling effort	• Add sampling intensity variables (n_samples/n_years) • Aggregate site level data • Minimum sampling threshold
2	**Correlated Predictors** Climate/geographic variable correlation	• Use of regularized regression for ridge and lasso • Compare correlation from lasso/ridge to tree models (tree models do not use coefficient estimation)
3	**Missing Environmental Data** Sites lack depth or climate data	• Restrict to sites with enough data (5+ years) • Use established climate proxies where needed (find temperature based on coordinates)
4	**Model Overfitting**	• Cross Validation (5 fold) • Parameter tuning • Model comparison

Alexandre H. Hirzel, Gwenaëlle Le Lay, Véronique Helfer, Christophe Randin, Antoine Guisan, Evaluating the ability of habitat suitability models to predict species presences, Ecological Modelling, Volume 199, Issue 2, 2006, Pages 142-152, ISSN 0304-3800, https://doi.org/10.1016/j.ecolmodel.2006.05.017.https://www.sciencedirect.com/science/article/pii/S0304380006002468

Dornelas et al. (2025). BioTIME 2.0: Expanding and Improving a Database of Biodiversity Time Series. Global Ecology and Biogeography, 34(5):e70003.

V. Braunisch, Conservation Biology, Inst. of Ecology and Evolution, Univ. of Bern, Baltzerstrasse 6, CH-3012 Bern, Switzerland. VB also at: Forest Research Inst. of Baden-Württemberg FVA, Wonnhaldestr. 4, DE-79100 Freiburg, Germanyhttp://climate.calcommons.org/bib/selecting-correlated-climate-variables-major-source-uncertainty-predicting-species-distributions#:~:text=Correlative%20species%20distribution%20models%20are,of%20the%20predictions%20of%20a).

Thank you!

Resources